Medical waste management

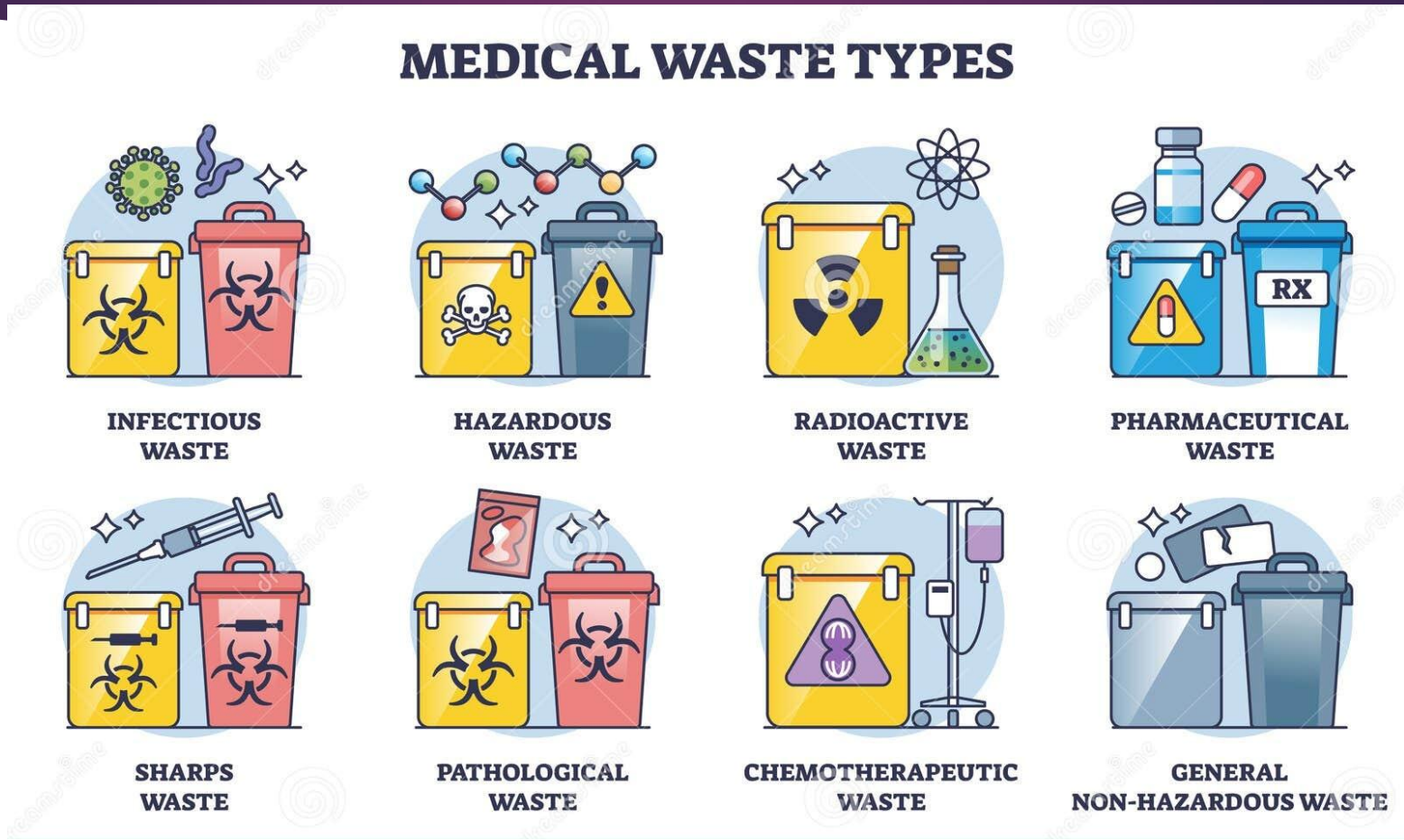
DR.EMAN.A.MAHMOUD

CLINICAL MICROBIOLOGY & IMMUNOLOGY

INFECTION CONTROL CONSULTANT

NLI

Types of medical waste:



Associated Diseases:

- **VIRUSES:** HIV, Hepatitis B, Hepatitis A,C& Ocular infection
- transmitted through Infected needles, body Fluids, Human excreta, Blood, body fluids, Eye secretion
- **BACTERIA:** Salmonella typhi, cholerae, Clostridium Tetani, Pseudomonas,, skin and soft tissue infections
- can be transmitted through Human excreta and body fluid in hospital wards, Sharps such as needles, surgical blades in hospital waste.
- **PARASITES**: Malaria, Cutaneous leishmaniasis, Filariasis,
- can be transmitted through Human excreta, blood and body fluids in poorly managed sewage system.

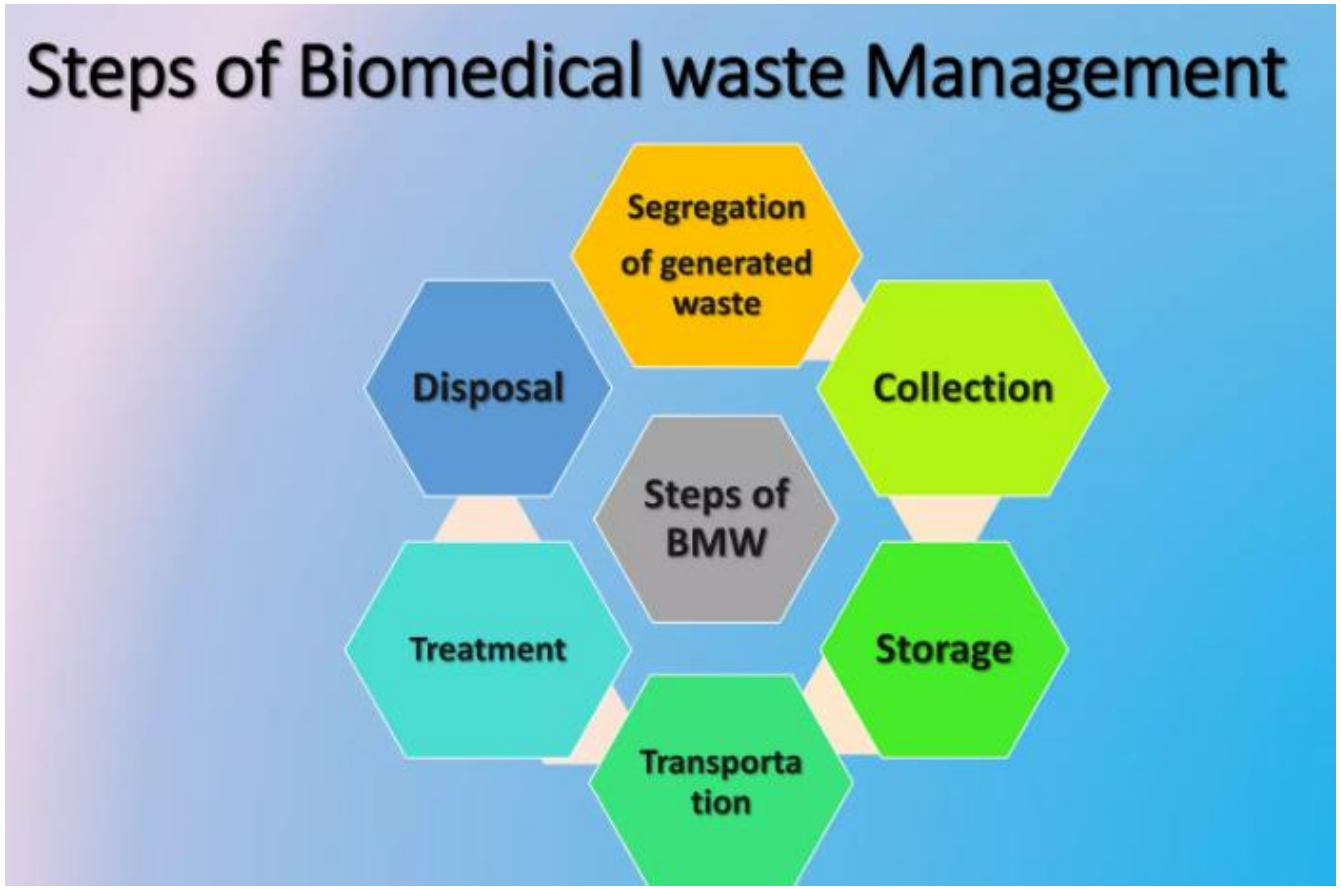
Radioactive Wastes

- **Produced as a result of:**
- in-vitro analysis of body tissue and fluid (PET using FDG)
- in-vivo organ imaging and tumor localization (PET/CT Fluorine-18)
- therapeutic practices (radiotherapy for tumor)
- Some have relatively **short half-lives** causing them to lose activity quickly (e.g., technitium-99m – 6 hours; iodine-123 – 13 hours; phosphorus-32 – 14 days)
- Others have long half-lives (carbon-14 – 5730 years)

Health Risks of Radioactive Waste

- **Health impacts determined by type and extent of exposure to ionizing radiation**
- Chromosomal damage
- Headaches, dizziness, vomiting
- Tissue destruction, hemorrhage, hair loss, diarrhea
- death at high doses

Medical waste management plan



separation /Sorting

- into color coded bags at the place where it is generated.
- should be at the point of origin by the individuals most knowledgeable about the waste's origin and its hazard potential
- Most errors in the medical waste management system occur at the segregation point

فصـــل المخلفـــات

لابد من فصل المخلفات عند المنبع أى فى مكان الإستخدام الذى تتكون فيه ويكون الفصل كالآتى :-

١- المخلفات العادية : المواد التى لم تتلوث بسوائل الجسم مثل:
المخلفات الناشئة عن المطبخ – الورق – الصناديق – مواد تغليف المستلزمات – الزجاجات والأوانى البلاستيكية – المناديل مخلفات الطعام.

ويتم القاء ذلك النوع من المخلفات فى ← **الكيس الأسود**



٢- المخلفات المعدية : المواد التى تلوثت أو يشتبه تلوثها بسوائل الجسم مثل:
أكياس الدم ومشتقاته – بعض الواقيات الشخصية بعد استعمالها (القفازات بأنواعها-الماسك-الأبرون البلاستيك) – أجهزة الوريد – القساطر (وريدية-بولية) – المزارع الميكروبية – نواتج الغيار (الشاش-القطن-الضمادات-المسحات-الويرات-الغرز الجراحية)

ويتم القاء ذلك النوع من المخلفات فى ← **الكيس الأحمر**



٣- المخلفات الحادة : كل ماهو حاد ويمكن أن يسبب خدش أو اختراق للجلد مثل:
السرنجات – السنون – الأبر – المشارط – الشكاكات – الفيالات الخاصة بالأدوية والمضادات الحيوية – أنابيب الاختبار.

ويتم القاء ذلك النوع من المخلفات فى ← **صندوق الأمان**



separation of Radioactive Waste

- **based on :**
- • Half-life
- • Activity and radionuclide content
- • Physical and chemical form –

Containers and labeling

On every label

Isotope, activity, date stored, initials and dose rate

Radiation symbol

Visible trefoil on container and bag

Separate by half-life

Short-lived waste never mixed with long-lived waste

Lids and seals

Closed containers; no loose waste



II. Handling

- is collecting and transporting waste within the facility.
- Staff involved in handling should be vaccinated against HBV& tetanus.
- Housekeepers should wear heavy utility gloves and eye protection during cleaning of the containers.
- Plastic bags Should be impervious, tear resistant
- Sharps containers should be puncture-resistant, leak proof on the sides and bottom, and closable.
- containers should be discarded when they are 2/3 full or at least once per day; sealed, lifted by the top.



III. Transport of medical waste

- **three types :**
  - **Collection from the immediate generating** area and transport to the temporary storage area on the generating unit.
  - **Transport from the temporary storage** area on the generating unit to the long term central collection and storage area.
  - **Transport from the facility** to a treatment or disposal site.
  - Transportation should be done in sealed container/sanitation supervisor should ensure for leakage.

IV. Interim storage:

- Interim storage is storing waste within the facility until it can be transported for final disposal.
- Waste should not be stored in the generation area for more than a period of 48 hours

The waste storage room

The room

- Locked, restricted access
- Radiation trefoil sign on door
- Shielded walls and floor as required
- Ventilated and fire-resistant

Inside the room

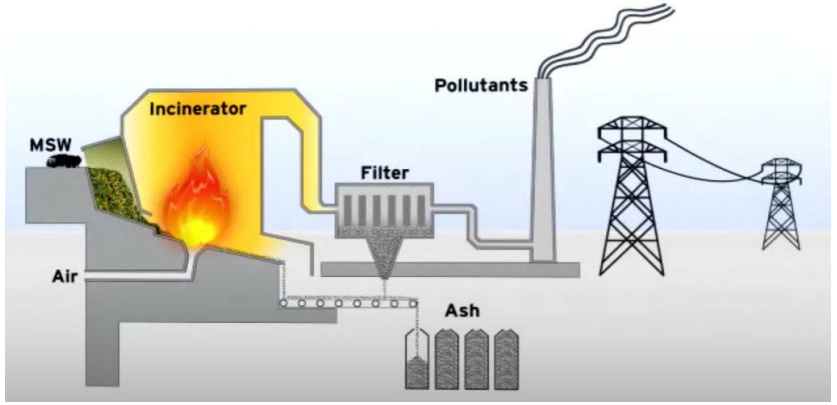
- Shielded, labeled containers
- Separate bins by isotope and half-life
- Spill tray under liquid waste
- Survey meter kept nearby

V. Treatment & Disposal

- **General waste** should be dumped at municipal dumping site. Sanitation officer should be responsible for proper coordination between municipal and hospital
- **Other wastes** managed by **different technologies** according to their types
- **Treatment technologies**
- 1.Incineration
- 2.Chemical Disinfection
- 3.Wet and dry thermal treatment
- 4.Microwave irradiation
- 5.Land disposal
- 6. Decay in Storage

Incineration

- Incineration is a common method of commercial disposal typically used for pathological and pharmaceutical waste.
- a high-temperature process that reduces organic and combustible waste to inorganic, incombustible matter and results in a very significant reduction of waste volume and weight
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Chemical Disinfection

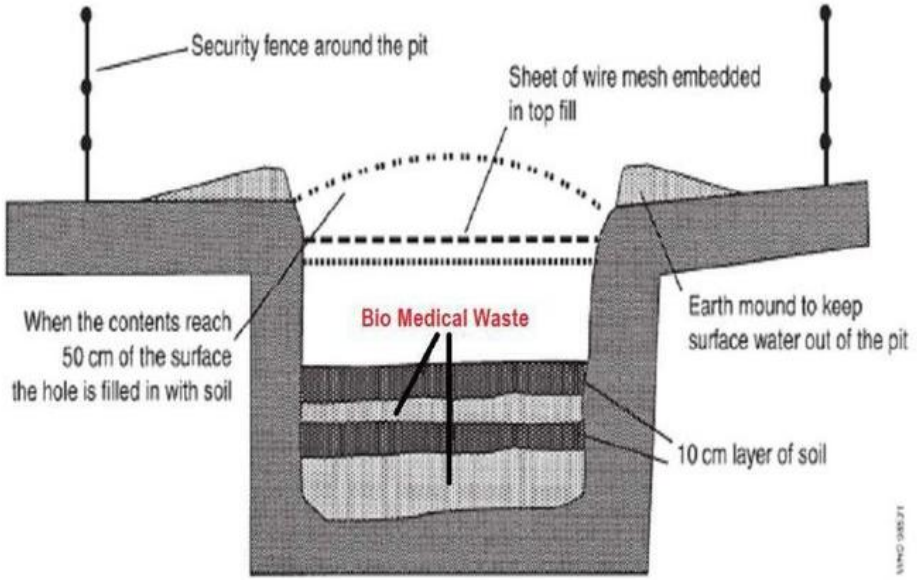
- This process involves the use of chemical agents for disinfection, such as chlorine,
- . Chemical disinfection processes are most appropriate for liquid wastes, although they can still be used to treat solid wastes.

thermal treatment

- **Autoclave:**
- In this method, wastes are sterilized or disinfected prior to disposal in a landfill.
- Bags of waste are placed in a chamber and steam is introduced for a determined period of time at a specified pressure and temperature. This assures the destruction of microorganisms.
- 90% of regulated medical wastes are suitable for autoclaving.
- Autoclaves are not suitable,, for pathological, cytotoxic, or other toxic chemical wastes.

Land disposal

- method of disposing of solid waste on land without creating hazards to public health or safety,



Decay in Storage

For radio active waste

- Store the waste for at least 10 times the half-life of the longest lived radionuclide in the waste
- Conduct a radiation survey to confirm that radioactivity is below the clearance level
- Disinfect mixed radioactive infectious waste before discarding with regular waste
- Long-term storage at an authorized radioactive waste disposal facility

Steralization & disinfection

definitions

- **Disinfection:** Process by which we can minimize the number of microorganisms except bacterial spores.
- **Steralization:**the process of eliminating all forms of microorganisms, including bacteria, viruses, fungi, and spores, from medical instruments, equipment, and surfaces
- **Disinfectant:** chemical substance to be used to disinfect (Inanimate) surface.
- **Antiseptic:** Chemical substance can be used to disinfect (animate) surface.

Levels of disinfection

High disinfection

- **Effectiveness**: Kills most microorganisms, including bacteria, viruses, fungi, and spores, but not necessarily all spores (some highly resistant spores may survive).
- **Organisms Targeted**: Includes Mycobacterium tuberculosis, all fungi& Hepatitis B virus
- **Disinfectant Examples**:

Glutaraldehyde

Hydrogen peroxide

Peracetic acid

Intermediate Disinfection

- **Effectiveness**: Kills most bacteria, viruses, and fungi, but not bacterial spores
- **Organisms Targeted**: Includes Staphylococcus aureus, Escherichia coli,HCV, influenza virus and fungi.
- **Disinfectant Examples**:

Alcohol (70%-90%)

Iodophors ex betadine

Low Disinfection

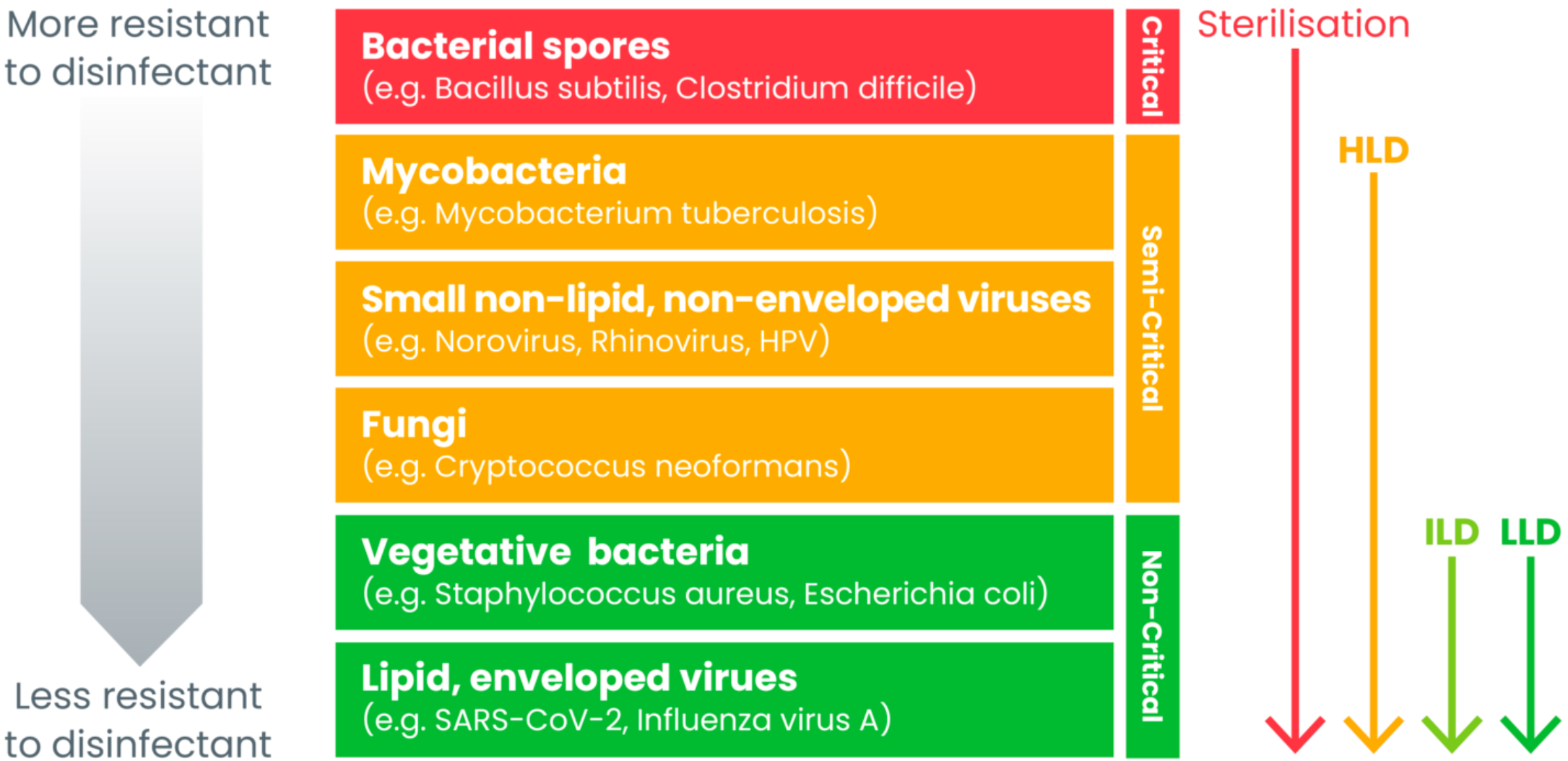
- **Effectiveness**: Kills most bacteria and some viruses, but not resistant organisms like Mycobacterium tuberculosis or bacterial spores.
- **Organisms Targeted**: Includes Streptococcus, Staphylococcus aureus, Hepatitis A, some fungi and ,HIV
- **Disinfectant Examples**:

Quaternary ammonium compounds(Sterillium ,cavicid)

Phenolic compounds (phenic,dettol)

Soap and water

Indicates What Level of Disinfection Eliminates all Microbe Groups[1-4]



Spaudling classification

Critical

Touches

Sterile tissue / bloodstream

Requires

Sterilization

Semi-critical

Touches

Mucous membranes, non-intact skin

Requires

High-level disinfection

Non-critical

Touches

Intact skin only

Requires

Low / intermediate-level

Critical items

Radiology examples

- Biopsy needles under CT / US
- Angiography catheters, guidewires
- Interventional radiology instruments
- Drainage catheters
- Contrast syringes and tubing to the patient

Required level

Sterilization (or sterile single-use)

Semi-critical items

Radiology examples

- Transvaginal / transrectal US probe
- HSG cannula and instruments
- Rectal tip in barium enema
- US probe used on an open wound

Required level

High-level disinfection (minimum)

A probe cover does not change the class. Covers can leak: disinfect after every patient.

Non-critical items

Radiology examples

- Abdominal US probe on intact skin
- Examination table, CT gantry
- MRI coils, lead aprons, cassettes
- Control panels, injector surface

Required level

Low or intermediate-level disinfection

High-touch surfaces spread infection via hands. Blood spill: raise to intermediate level.